MAE 3270

Fall 2025
Torque Wrench Design

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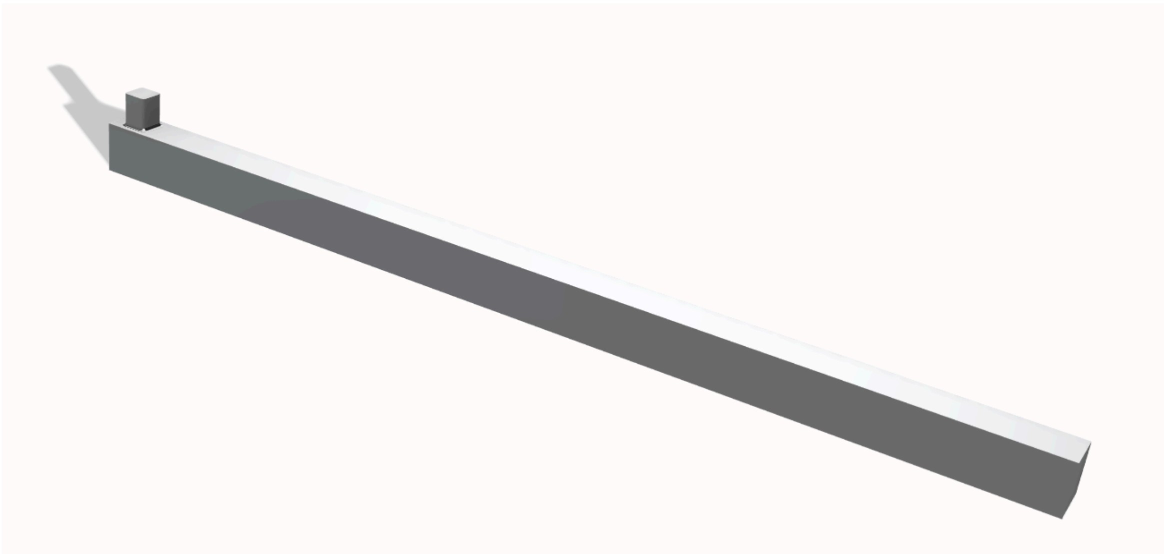
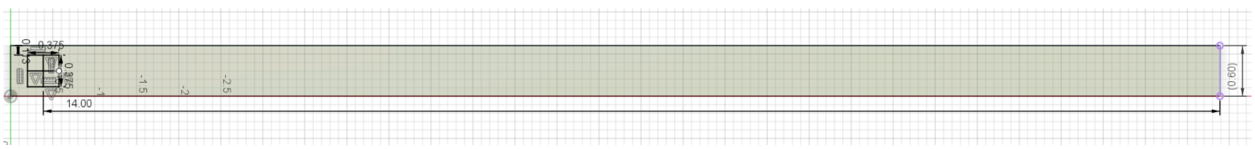
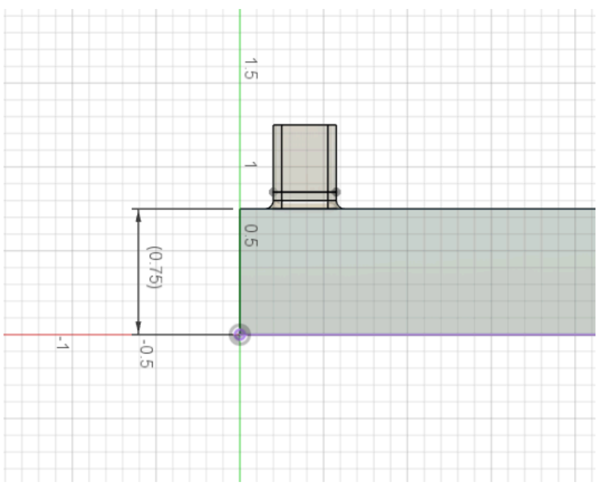
1. Image of CAD models



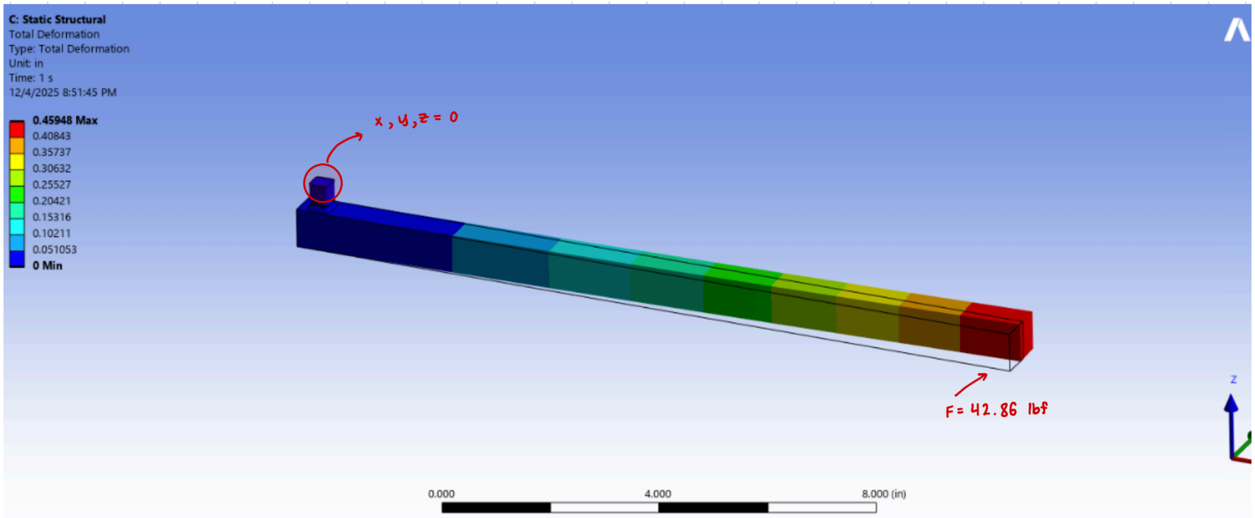




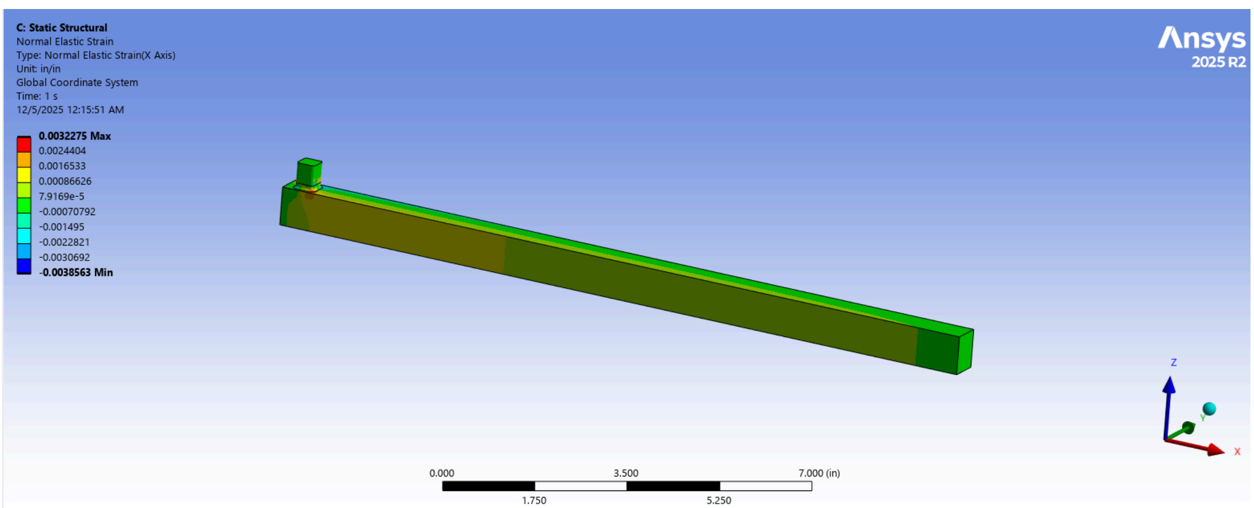
Main dimensions:
L = 14in
b = 0.75 in
h = 0.6 in
c = 1 in

2. For our design we used aluminum 7075-T651 as it has high strength while being light which is great for lightweight and high performance applications. With a tensile strength of 70,000 psi, it can withstand significant mechanical loads without substantial deformation. In addition, its lower density compared to steel helps reduce overall weight, resulting in a lighter, more ergonomic design that is easier for users to handle.

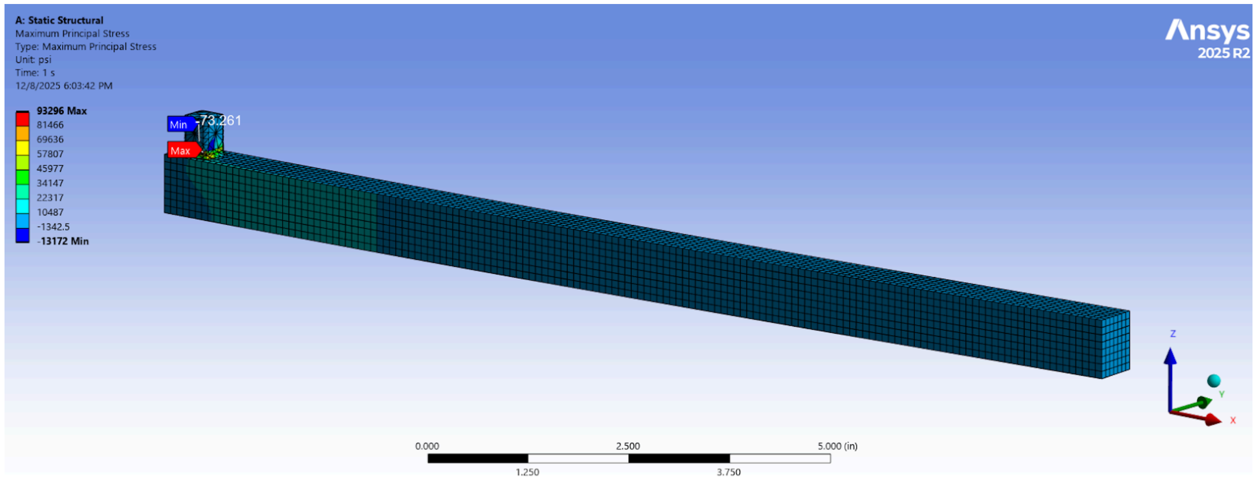
3. We held the drive end at constant by setting the displacement components equal to 0. Then we applied 42.86lbf = Torque/Length at the end face in y direction.



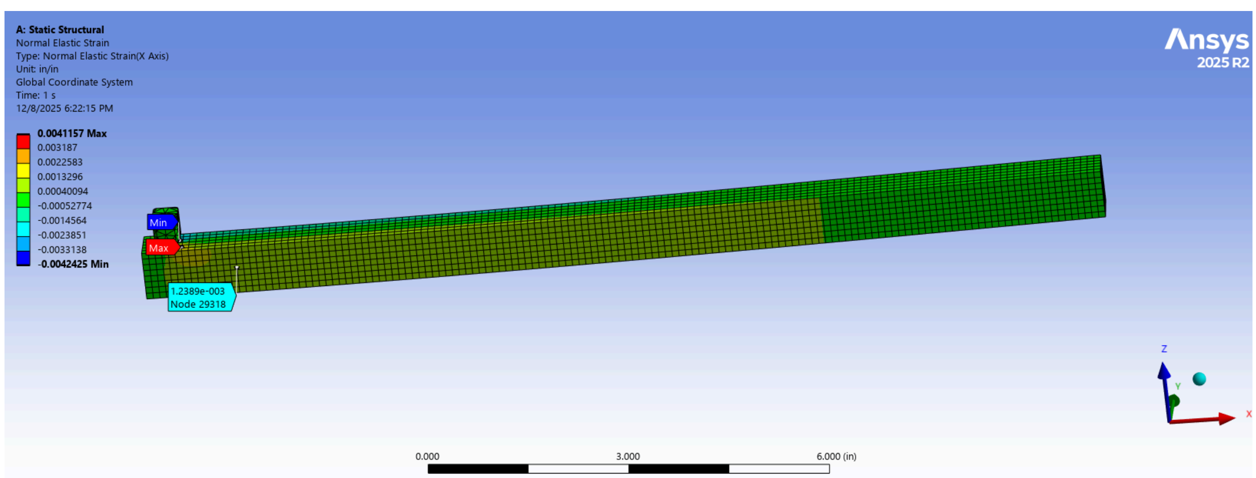
4. Normal strain contour



5. Contour plot of maximum principle stress



6. We found that the maximum normal stress in the structure reaches 41,400 psi, while the maximum deflection at the load point is 0.459 inches. At the strain gauge location, the FEM predicts a stress of 12,690 psi, which aligns closely with our MATLAB calculation of 13,333 psi.



7.

$\frac{V_{out}}{V_{in}} = \frac{2k\epsilon}{4}$, gauge factor K = 2, strain at strain gauge = 1.238e-3

mV/V = 1000*strain = 1000* 1.238e-3 = **1.238**

8. SGD- 2/350-LY11
    a. Normal Resistance: 350 omega
    b. Gauger factor = 2
    c. active grid length: 2mm (0.118 in)
    d. active grid width: 2.5mm (0.098 in)
    e. Matrix length: 7.6 mm (0.299 in)
    f. Matrix width: 5.8 mm (0.228 in)

Our design cross section where the gauge will be applied is 0.75x0.6 in, providing us with enough space for the gauges to bond.